

Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN

KEIL AIM:

To write and simulate an Embedded C program using serial interrupt in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from

the keyboard generates an interrupt and is displayed through UART.

APPARATUS / SOFTWARE REQUIRED:

- PC with Windows OS
- Keil μ Vision IDE
- 8051 Simulator (Keil built-in)
- Serial Window (UART #1)

PROGRAM:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
void serial_ISR(void) interrupt 4
```

```
{
```

```
char ch;
```

```
if (RI)
```

```
{
```

```
RI = 0; // Clear receive interrupt flag ch =
```

```
SBUF; // Read received character
```

```
printf("&quot;\n&gt;&gt;&gt; INTERRUPT RECEIVED: &quot;);
```

```
printf("&quot;%c\n&quot;;, ch);
```

```
}
```

```
}
```

```
void delay(void)
```

```
{
```

```
    unsigned int i;
```

```
    for(i = 0; i < 50000; i++); // Delay for visibility }
```

```
void main(void)
```

```
{
```

```
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
```

```
    TMOD = 0x20; // Timer1 mode 2 (auto-reload) TH1 =
```

```
    0xFD; // Baud rate 9600
```

```
    TR1 = 1; // Start Timer1
```

```
    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt) TI
```

```
    = 1; // Enable printf
```

```
    while(1)
```

```
    {
```

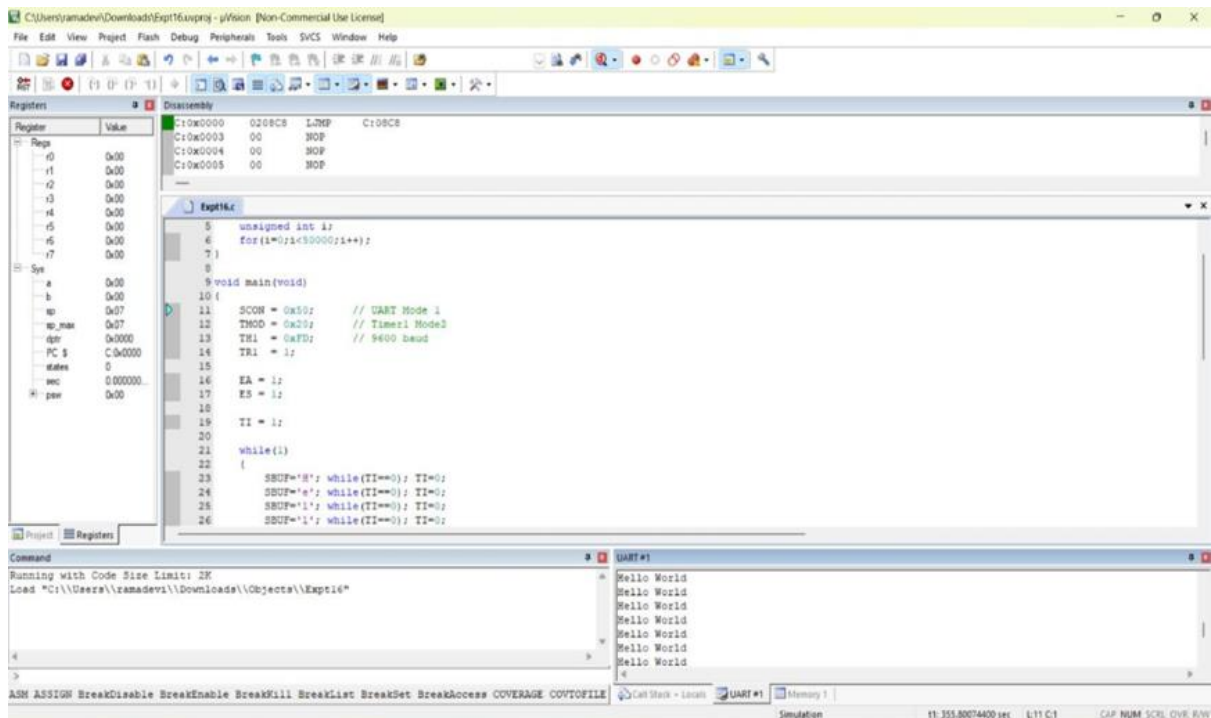
```
        printf("Hello World\n");
```

```
        delay();
```

```
    }
```

```
}
```

OUTPUT:



RESULT:

Thus, the serial interrupt using UART was successfully implemented and verified in Keil simulator. The interrupt was triggered upon keyboard input and serviced correctly using an Interrupt Service Routine.